A Logarithm Answers an Exponent Question

$a^u = a^v \iff u = v$

For $a > 0$, $a \neq 1$, and $x > 0$,

$$y = \log_a x \iff a^y = x.$$

Evaluate without a calculator: $\log_2 8$, $\log_5\left(\frac{1}{25}\right)$, and $\log_3 1$. For each answer, write the matching exponential statement.

- In order for $\log_a x$ to be a valid function, we need $a > 0$, $a \neq 1$, $x > 0$.

- $f(x) = \log_a(x)$

  $y = \log_a(x) \iff a^y = x$

- $\log_2(8) = y \iff 2^y = 8$

  $2^y = 2^3$

  $y = 3$

$\log_5\left(\frac{1}{25}\right) = y$

$5^y = \frac{1}{25}$

$5^y = 25^{-1} = (5^2)^{-1}$

$5^y = 5^{-2}$

$y = -2$

- $\log_3 1 = y$

  $3^y = 1$

  $y = 0$

Translate Between the Two Forms

$\log_a(x) = y \iff a^y = x$

Rewrite each statement in the other form. Do not approximate. $e^m = 7$

(a) $3^{-4} = \frac{1}{81}$ (b) ~~$e^m = 7$~~ (c) $\log_2 64 = 6$.

Then evaluate $\log_4 64$ and $\log_8 64$ by asking which exponent is needed.

(a) $3^{-4} = \frac{1}{81}$

$a^y = x$ form

$a=3,\ y=-4,\ x=\frac{1}{81}$

$\log_3\left(\frac{1}{81}\right) = -4$ ✓

(b) $e^m = 7$

$a=e,\ y=m,\ x=7$

$\log_e(7) = m$

$\ln(7) = m$ ✓

(c) $a=2,\ x=64$

$y=6$

$2^6 = 64$ ✓

Solve a Logarithmic Equation

$a>0, a\neq 1$ argument is x/input

$x>0$

Solve each equation. Check the logarithm's argument and base restrictions.

(a) $\log_2(3x-1)=3$ (b) $\log_a(16)=2$.

$\log_a x=y \Leftrightarrow a^y=x$

$a>0, a\neq 1$

(a) base is 2.

$2>0, 2\neq 2$ ✓

$3x-1>0$

$3x>1$

$x>\frac{1}{3}$

$\log_2(3x-2)=3$

$2^3=3x-2$

$8+2=3x$

$x=9$ ✓

since $9>\frac{1}{3} \Leftrightarrow x>\frac{1}{3}$

so $x=9$ is valid.

(b) $\sqrt{a^2}=\sqrt{16}$

$a=\pm 4$

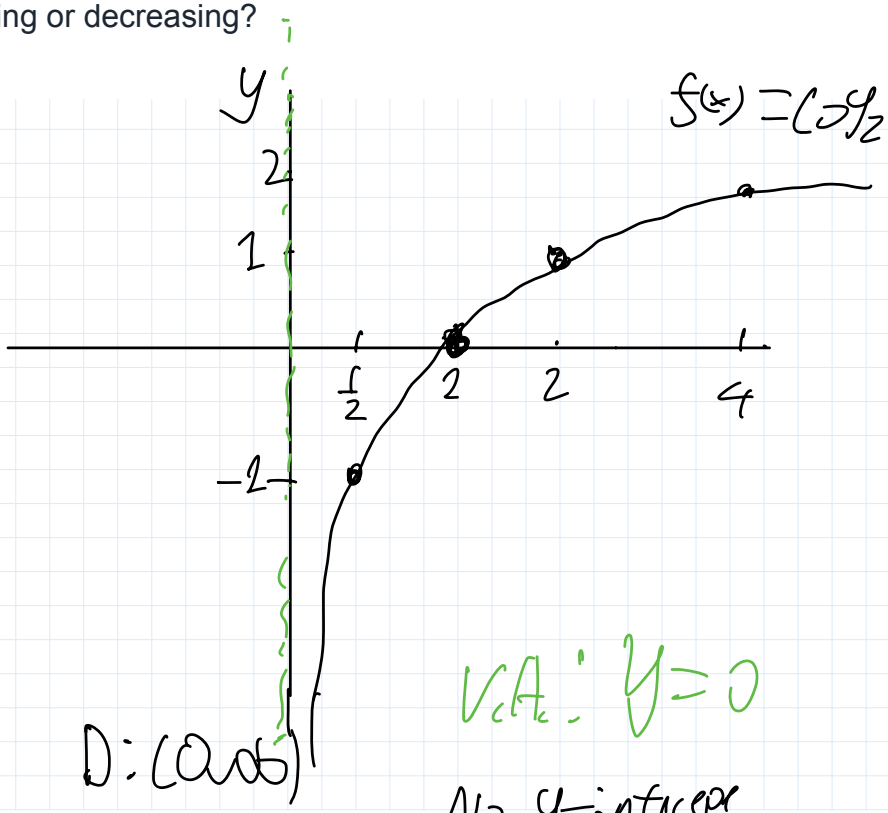
$a=-4$ is not valid.

so we left with $a=4$.

The Basic Logarithmic Graph

The functions $y = \log_2 x$ and $y = 2^x$ are inverses. Use points on 2^x to sketch $y = \log_2 x$. State its domain, range, vertical asymptote, and intercepts. Would $y = \log_{\frac{1}{2}} x$ be increasing or decreasing?

2^x	$\log_2(x)$
$(-1, \frac{1}{2})$	$(\frac{1}{2}, -2)$
$(0, 1)$	$(2, 0)$
$(1, 2)$	$(2, 1)$
$(2, 4)$	$(4, 2)$



$f(x) = \log_2(x)$

$\log_2 x = 0$

$2^0 = x$

$x = 1$

$(1, 0)$

V.A.: $y = 0$

D: $(0, \infty)$

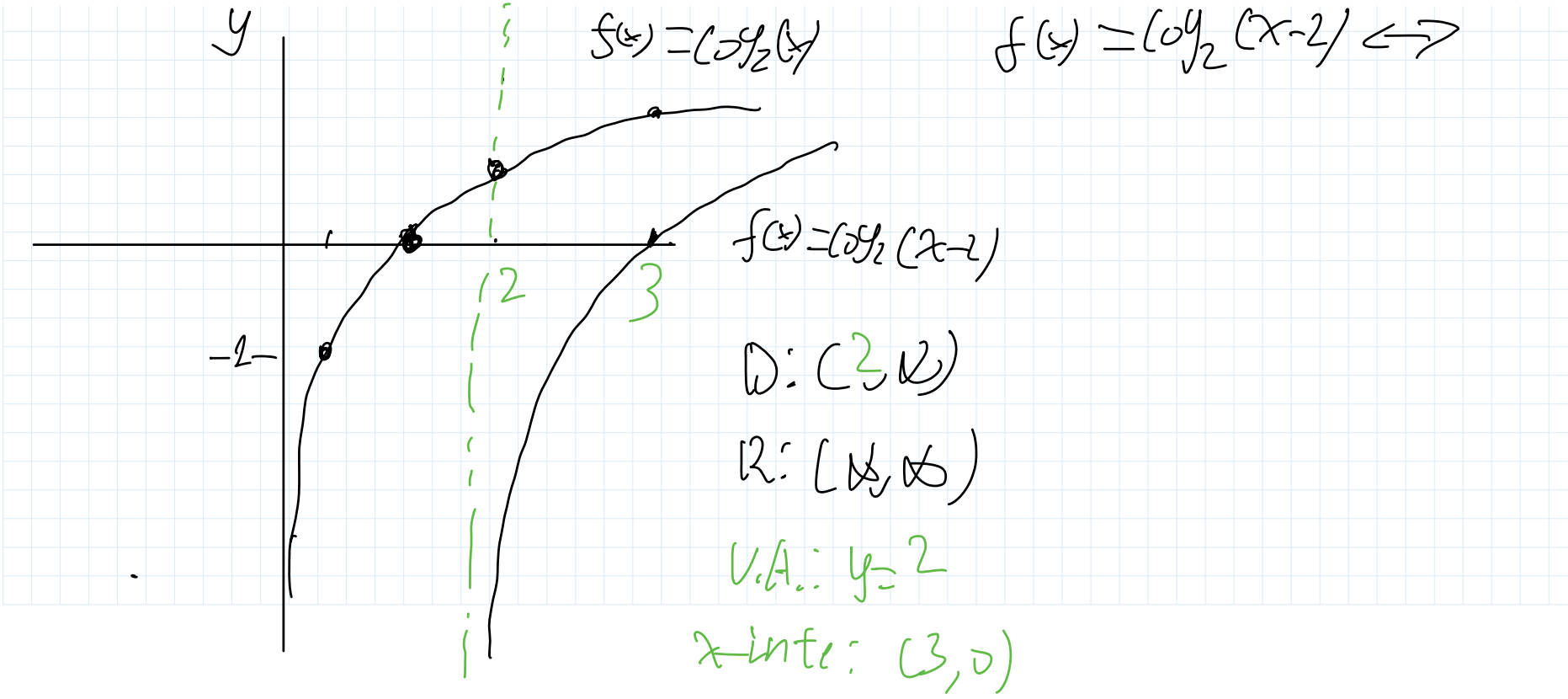
R: $(-\infty, \infty)$

No y-intercept.

x-int: $(1, 0)$

A Shifted Logarithmic Graph

Sketch $f(x) = \log_2(x-2)$. State the domain, range, vertical asymptote, and intercepts. Explain the shift from $y = \log_2 x$.



$y = \log_a x \Leftrightarrow a^y = x$

Natural and Common Logarithms

Natural log: $\ln x = \log_e x$. **Common log:** $\log x = \log_{10} x$. Evaluate exactly: $\ln(e^3)$, $\log(0.01)$, and $e^{\ln 5}$. Then state the domain of $h(x) = \ln(4 - x)$. Is $\ln(-2)$ a real number?

- $\log_e x = \ln x$
- $\log_{10} x = \log x$

ex: $\log_4(4^6) = 6$

$\log_e(e^5) = 5$

$\log_{10}(10^1) = 2$

$\ln(e^3)$ x

$\log_e(e^3) = y$

$y = 3$

- $\log(0.01) = y$ $\quad y = -2$

$e^y = e^3$

$10^y = 0.01$

$10^y = \frac{1}{100}$

$10^y = 100^{-2}$

$10^y = ((10)^2)^{-2}$

$10^y = 10^{-2}$

$e^{\ln 5} = 5$

$5 = e^{\ln 5} = f(f^{-1}(x))$

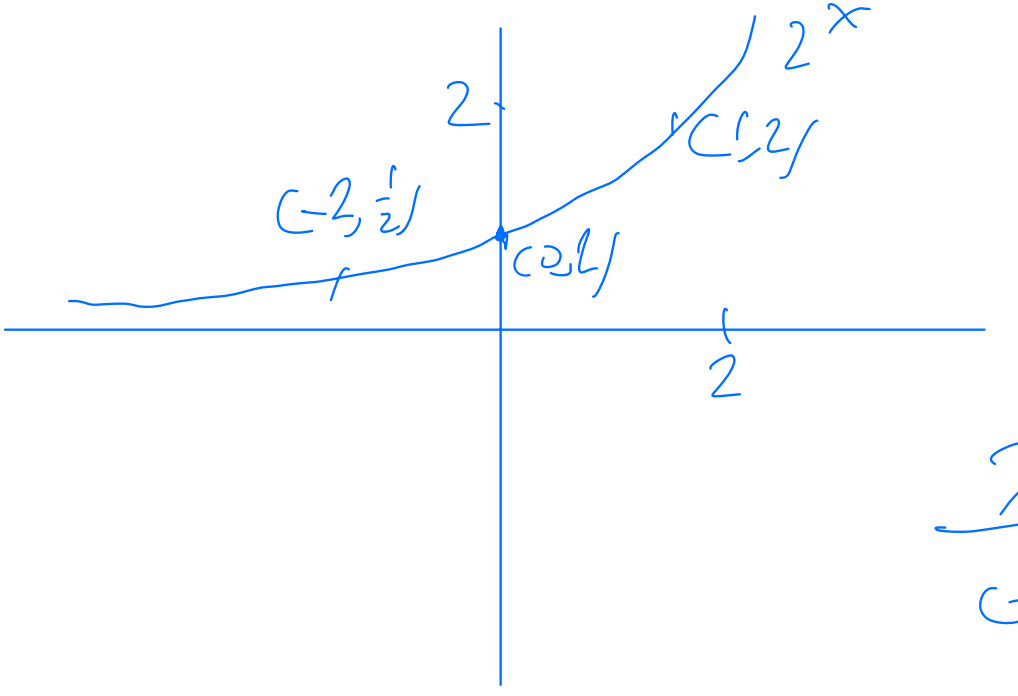
$= f(\ln x)$

$= e^{\ln x} = x$

$\begin{cases} f(x) = e^x \\ f^{-1}(x) = \log_e(x) = \ln(x) \end{cases}$

$\begin{cases} f(f^{-1}(x)) = x \\ f^{-2}(f(x)) = x \end{cases}$

$\begin{cases} f(x) = \sqrt{x} \\ f^{-1}(x) = x^2 \end{cases} \Rightarrow \begin{cases} f(f^{-1}(x)) = \sqrt{x^2} = x \\ f^{-1}(f(x)) = (\sqrt{x})^2 = x \end{cases}$



x	$y=2^x$
0	1
-1	$\frac{1}{2}$
1	2

$2^{x-1}-3$: right 1, down 3.